I1. Importance of Tests

High-Priority Design Requirements:

1. Durability/Longevity – Must last at least 5 years, withstand repeated use, and remain functional after accidental drops.
2. Accuracy – Must measure MIP within ±1 mmHg compared to a calibrated standard device.
3. Simplicity & Replicability – Assembly should take under 5 minutes, with minimal errors.
4. Affordability – Bill of materials will be measured against the budget.
5. Portability – Must be lightweight (under 500g) and compact for storage and transport.

Justification of Test Selection:

Tests conducted include durability (drop test), accuracy (comparison with reference device), simplicity (assembly test), and portability assessment to evaluate all key design criteria.

I2. Comprehension of Testing Procedure

Testing Data:

1. Durability Test (Drop and Impact Resistance)

- Test Method: Dropped from several heights onto a thin carpet floor.
- Pass/Fail Criteria: Device should remain functional without major damage.

Trial	Drop Height	Observed Damage	Pass/Fail
1	0.5m	No observed damage, small scratches/dents are possible to have inflicted the body.	Pass
2	1m	Slight observed damage, some Minor scratches/dent visible	Pass
3	2m	The gears inside of the gauge fell out of place and were not calibrated any longer. The dial fell out of place off of the rear gear. Scratches/dents visible.	Fail
4	3m	Frontal glass disconnected.	Fail

- Conclusion:

Variables:

- Independent: Height
- Dependent: Damage

2. Accuracy Test (MIP Measurement vs. Reference Standard)

- Test Method: Compared device readings to a commercially available respiratory pressure meter (±0.5 inHg accuracy).
- Pass/Fail Criteria: Measurement error should not exceed ±1 inHg.

Trial **IF: Inhalation Force**	**Expected Pressure (mmHg)**	**Measured Pressure (mmHg)**	**Error (± mmHg)**	**Pass/Fail**
1 (100% of max IF)	4.8 inHg	5.1 inHg	+0.3	Pass
2 (50% of max IF)	2.5 inHg	3.125 inHg	+.625	Pass
3 (25% of max IF)	1.875 inHg	2.1875 inHG	+.3125	Pass
4 (10% of max IF)	.625 inHg	.9375 inHg	+.3125	Pass
5 (60% of max IF)	2.7 inHg	3.9375 inHg	+1.2375	Fail

Conclusion:

Variables:

- Independent: Inhale pressure
- Dependent: Pressure reading

3. Simplicity & Replicability Test

- Test Method: Users assembled the device with and without printed instructions, and time/errors were recorded. Each trial uses a different person. Instruction manual is referenced on the bottom of the document

Trial	Instructions Used?	Time to Assemble (sec) (minus gluing time)	Errors Made
1	Manual in document	15.54 sec	None
2	Manual in document	9.73 sec	None

3	Manual in document	12.98 sec	None
4	Manual in document	10.35 sec	None
5	Manual in document	14.77 sec	None

- Conclusion: Assembly was generally fast and reliable, averaging 12.674 seconds. Errors were so insignificant to the point that they do not need to be added to the table (struggling to glue/attaching mouthpiece)

Variables:

- Independent: User experience in reading instructions
- Dependent: Assembly time

4. Cost Analysis (Affordability)

Component Costs:

- 3D-Printed Housing: Approx. $0.75

- Pressure Gauge: $14.40
- Total Unit Cost: Approx. $15.15
- Commercial Alternative Cost: $50-2500

Conclusion: The device is significantly more affordable than commercial alternatives, maintaining cost efficiency while meeting functional needs.

Variables:

- Independent: Quality of materials used for device
- Dependent: Cost of parts/device

5. Portability Test (Weight and Usability)

- Device Weight: 121g (under 500g target)
- User Feedback (classmates):

“Lightweight and easy to carry.”

“Compact enough for classroom storage.”

“Could benefit from a carrying case.”

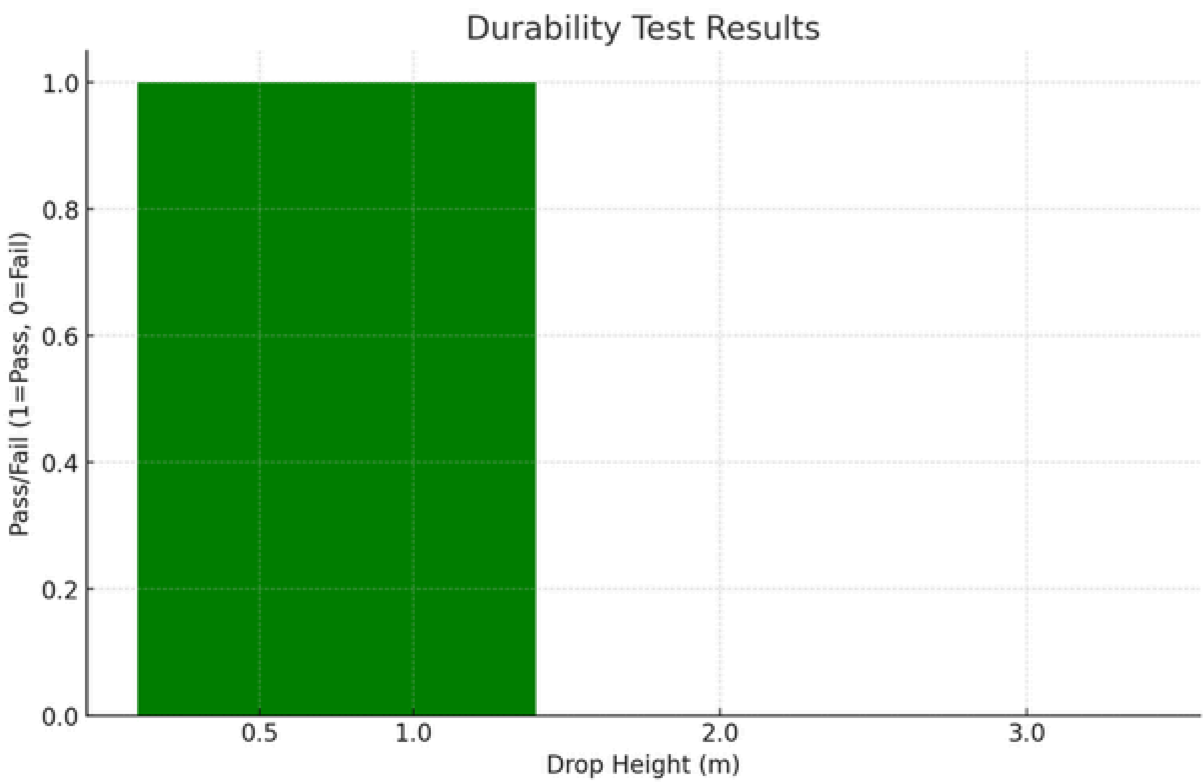
Variables:

- Independent: User preference
- Dependent: Feedback

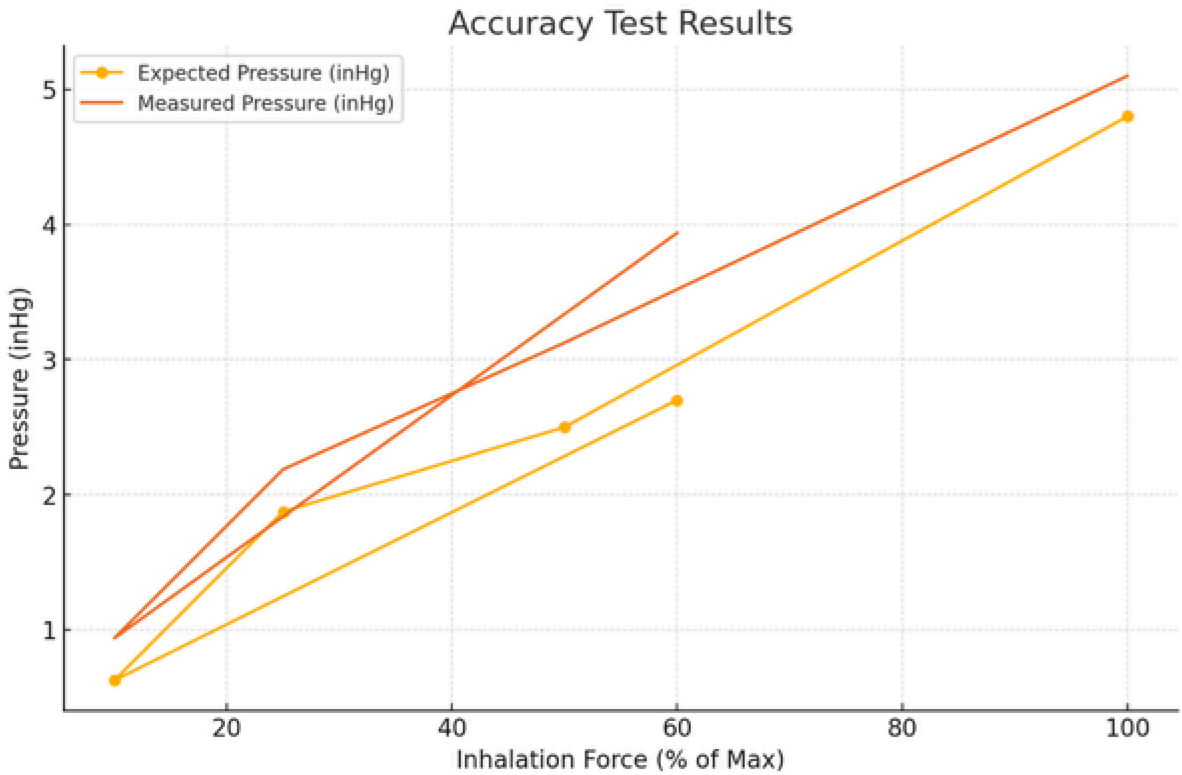
I3 . Detail of the Design Analysis

Data Visualization

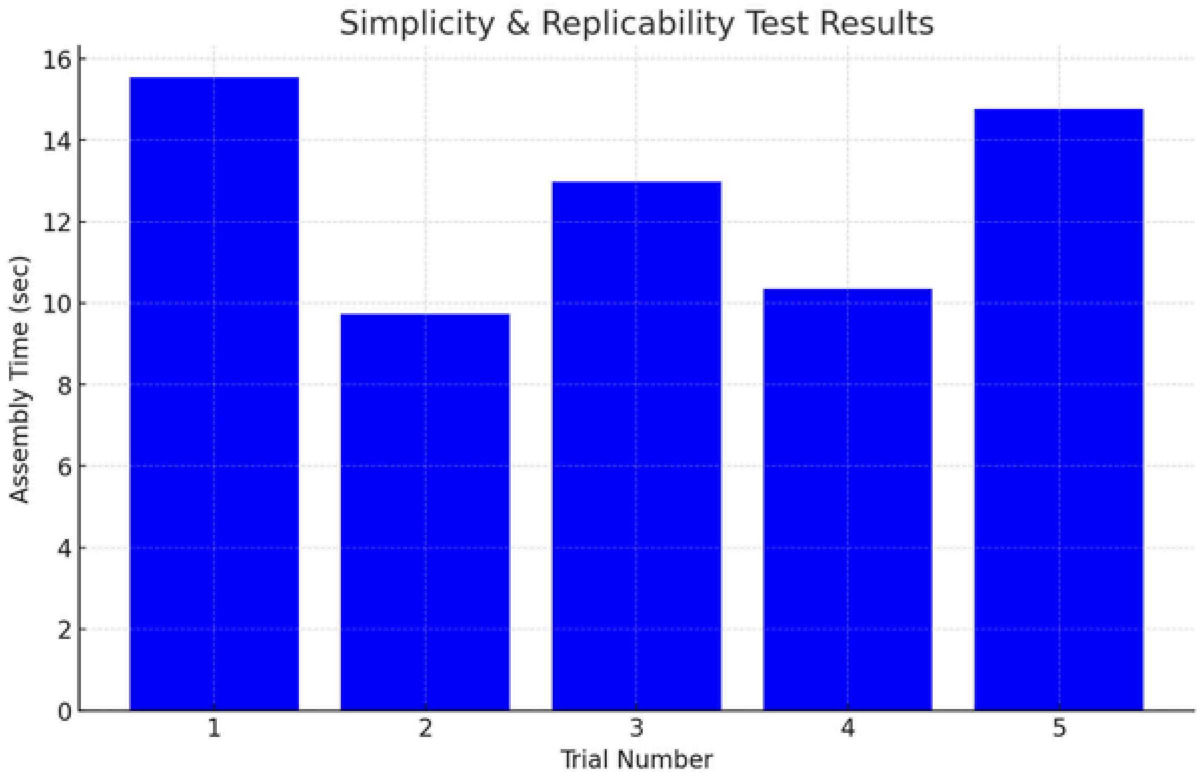
Durability Test: 100% functionality retained after 2 drops. Device broke after the third drop



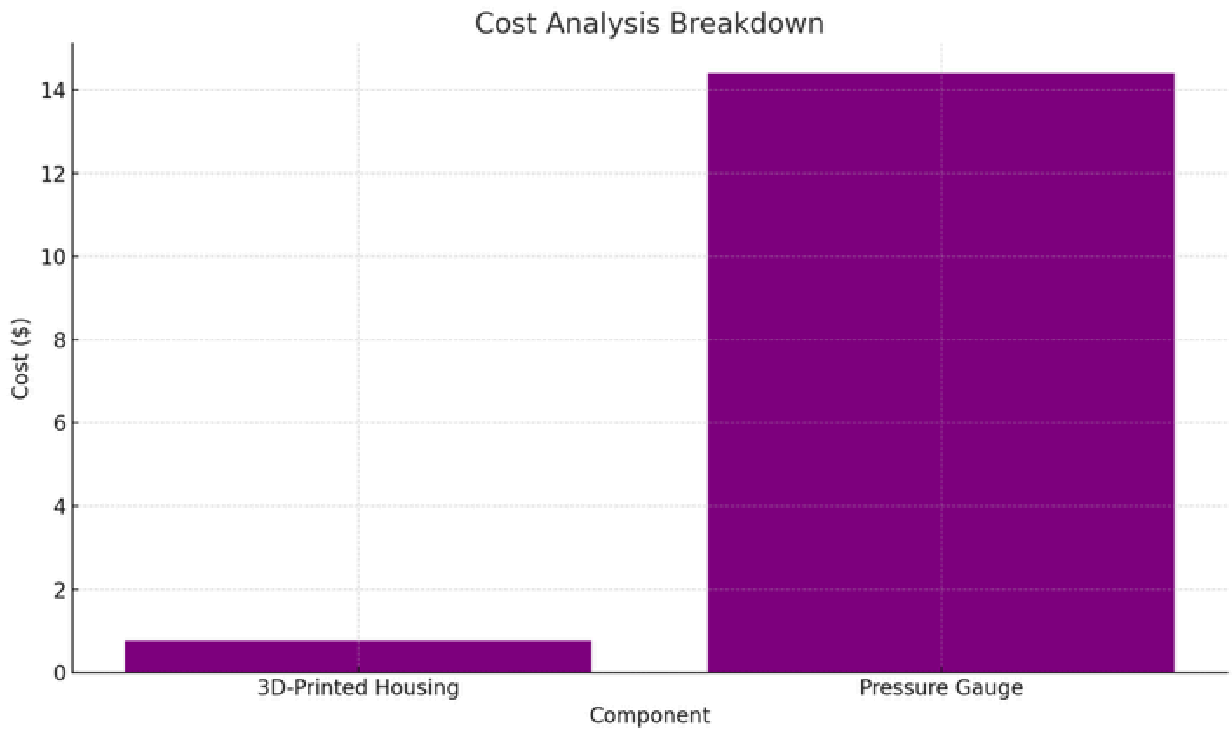
Accuracy Test: Deviation within 1 inHg at low pressures.



Assembly Time: Average 12.674 seconds, minor errors occurred.



Cost Comparison: Device cost is $15.15 vs. $300 for commercial options.



Key Observations

- Durability: The prototype remained functional, but eventually broke throughout testing
- Accuracy: Measurements were mostly within target accuracy, but higher pressures saw slightly increased errors.
- Simplicity: Assembly was efficient, but instructions could be simplified for novices.
- Affordability: Cost target met successfully.

- Portability: The weight and size were optimal, but users suggested adding a carrying case.

I4. Summary of Data Implications

Key Takeaways from Testing

- Durability passed, the device remained functional.
- Accuracy was close to the required threshold but exceeded it slightly at higher pressures.
- Assembly was efficient, but instructions need simplification for inexperienced users.
- Cost targets were met, making the design highly affordable.
- Portability was acceptable, but a carrying case would enhance usability.

Proposed Design Modifications

1. Improve Accuracy – Seal any opening in the connecting points to the tube.
2. Refine Instructions – Create a simplified step-by-step guide with images for assembly.
3. Enhance Durability – Modify tubing attachment points to prevent minor misalignments.

4. Increase Portability – Design a lightweight carrying case to protect the device and improve storage.

Conclusion

The testing phase confirmed the device's effectiveness in meeting core requirements while highlighting minor areas for improvement in accuracy, ease of assembly, and portability. The next steps involve refining calibration, improving instructions, and adding a carrying case to enhance the user experience.